

Image Segmentation

Image Segmentation

- Group similar components (such as, pixels in an image, image frames in a video)
- Applications: Finding tumors, veins, etc. in medical images, finding targets in satellite/aerial images, finding people in surveillance images, summarizing video, etc.

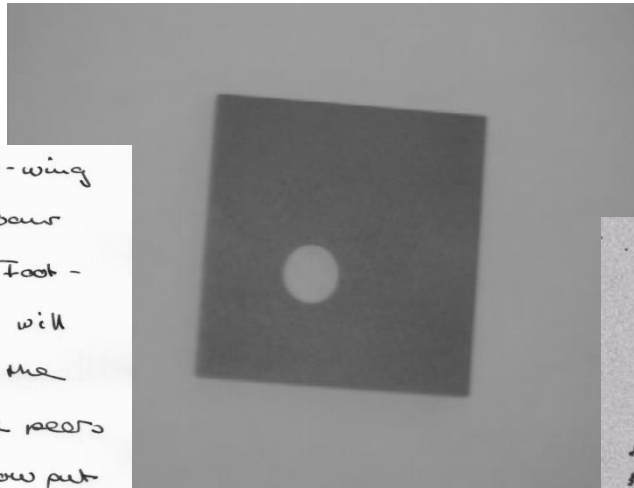
Image Segmentation

- Segmentation algorithms are based on one of two basic properties of gray-scale values:
 - Discontinuity
 - Partition an image based on abrupt changes in gray-scale levels.
 - Detection of isolated points, lines, and edges in an image.
 - Similarity
 - Thresholding, region growing, and region splitting/merging.

Thresholding

- Segmentation into two classes/groups
 - Foreground (Objects)
 - Background

Though they may gather some left-wing support, a large majority of Labour MPs are likely to turn down the Foot-Griffiths resolution. Mr. Foot's line will be that as Labour MPs opposed the Government Bill which brought life peers into existence, they should not now put forward nominees. He believes that the House of Lords should be abolished and that Labour should not take any steps which would appear to "prop up" an out-



Thresholding

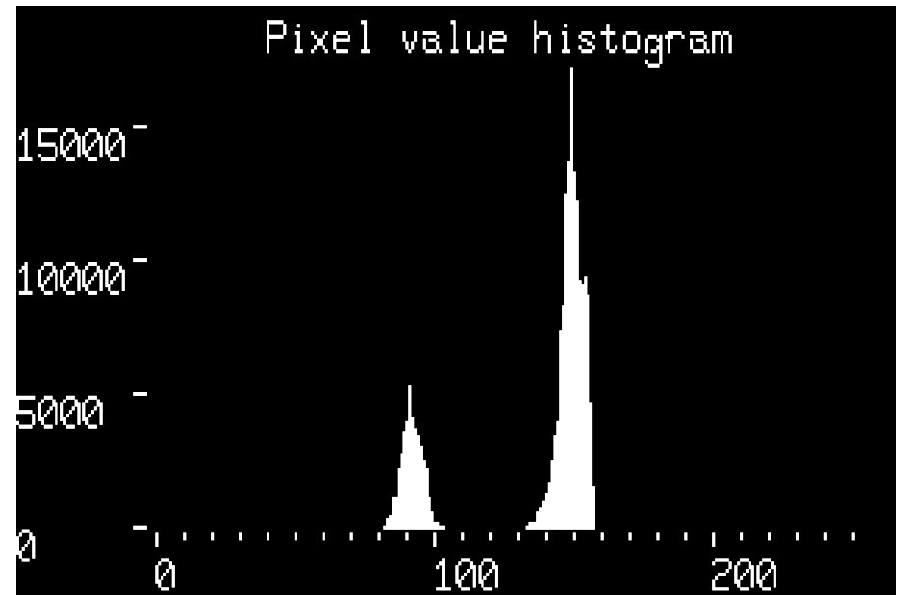
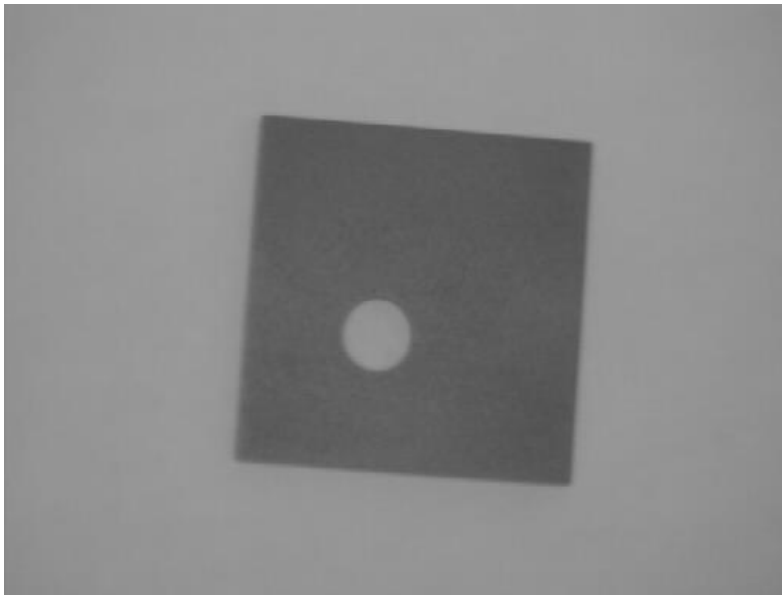
$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

Objects & Background

- Global Thresholding
- Local/Adaptive Thresholding

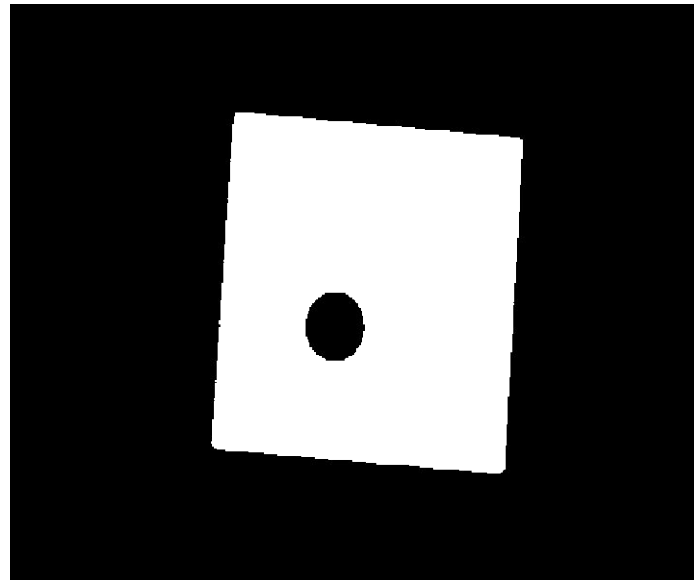
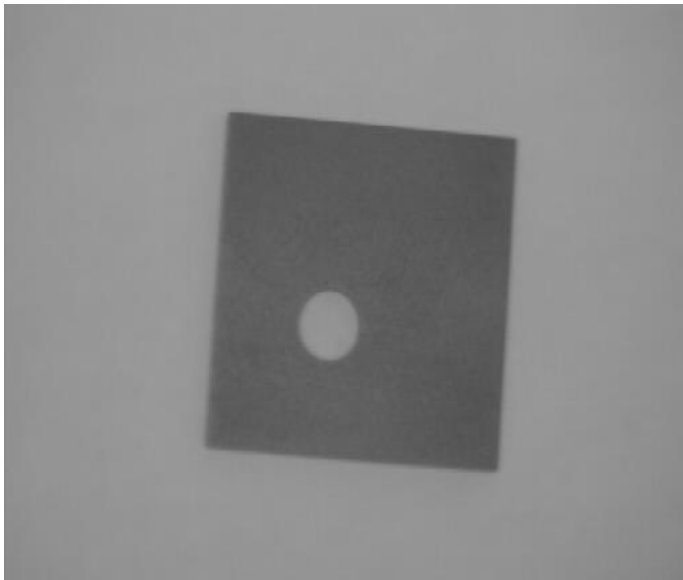
Global Thresholding

- Single threshold value for entire image
- Fixed ?
- Automatic
 - Intensity histogram



Global Thresholding

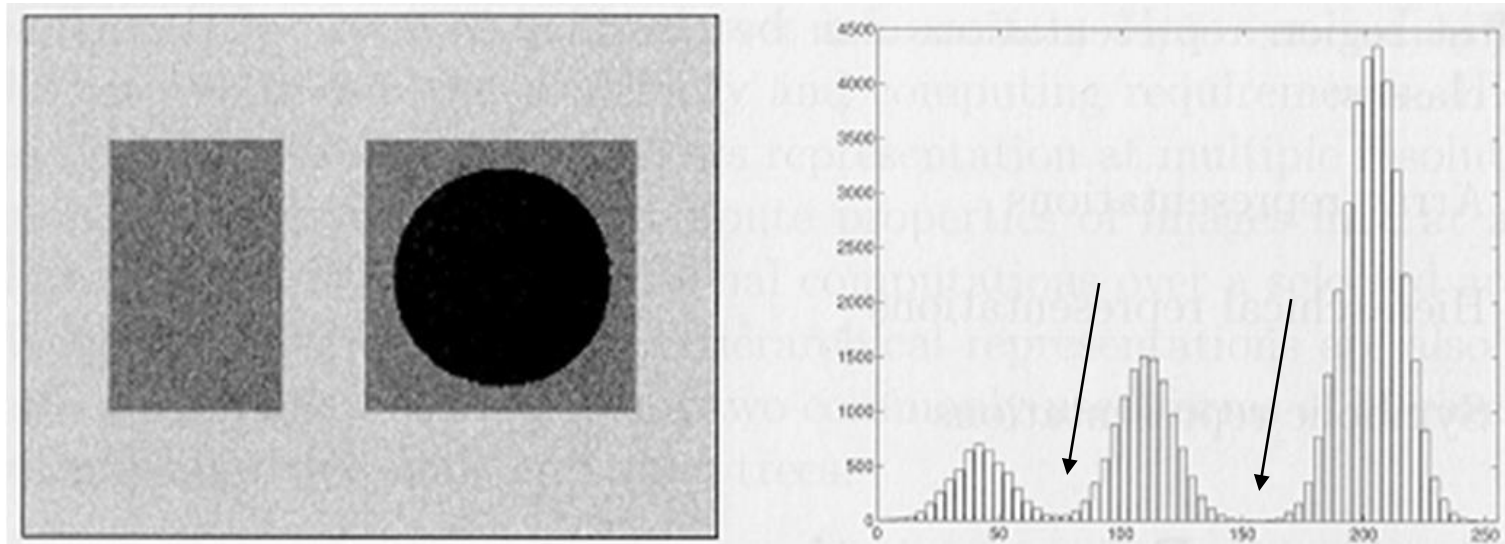
- Single threshold value for entire image
- Fixed ?
- Automatic
 - Intensity histogram



Global Thresholding

- Estimate an initial T
- Segment Image using T : Two groups of pixels $G1$ and $G2$
- Compute average gray values $m1$ and $m2$ of two groups
- Compute new threshold value $T = 1/2(m1 + m2)$
- Repeat steps 2 to 4 until: $\text{abs}(T_i - T_{i-1}) < \text{epsilon}$

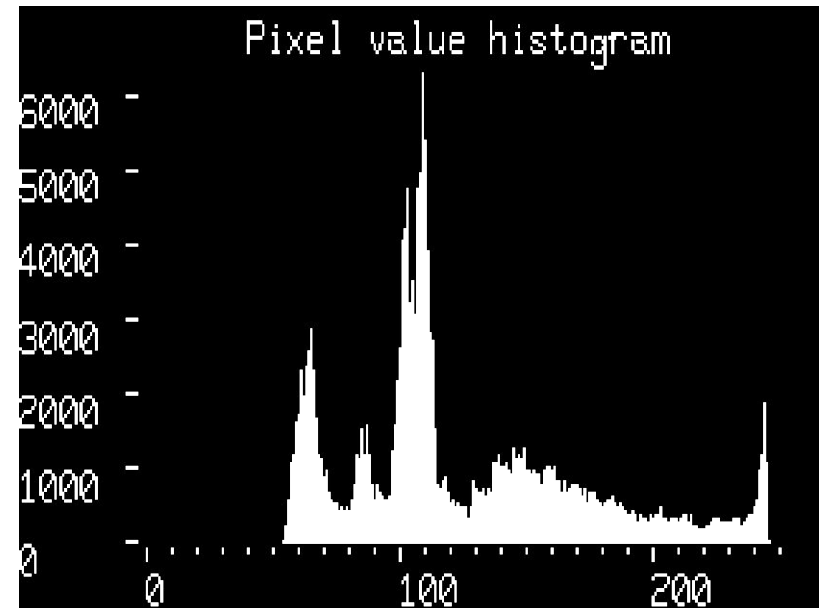
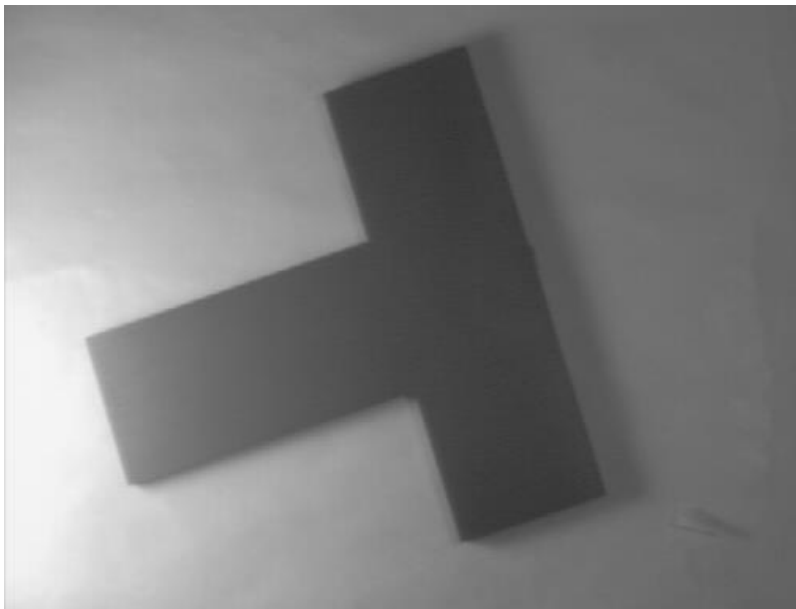
Global Thresholding



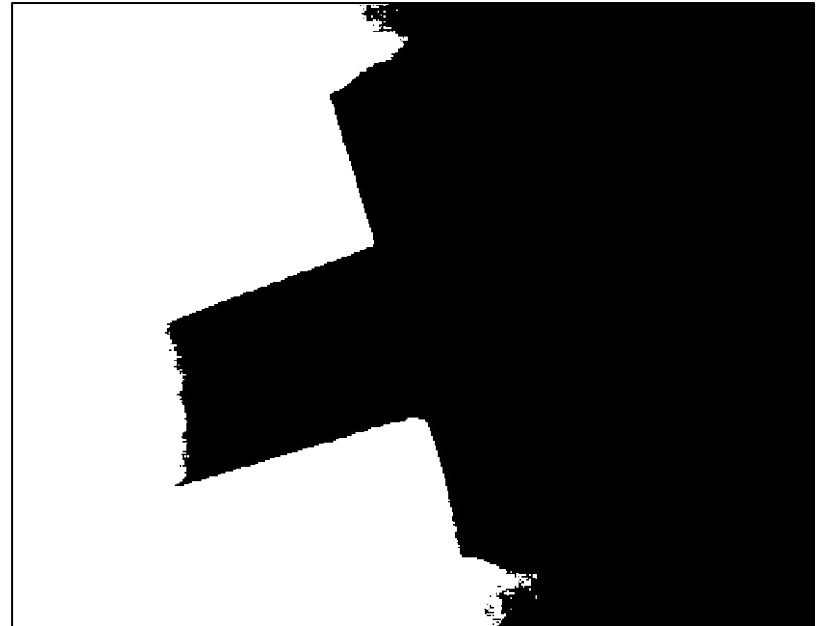
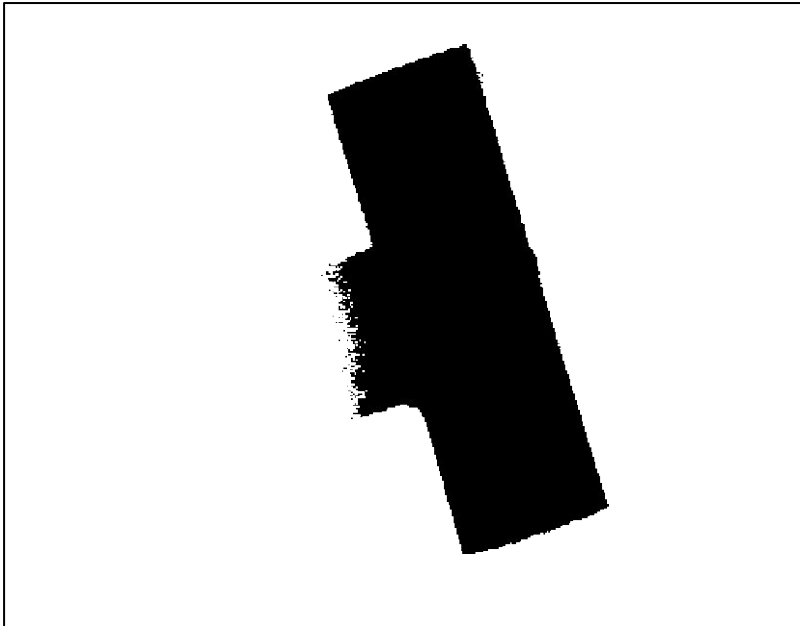
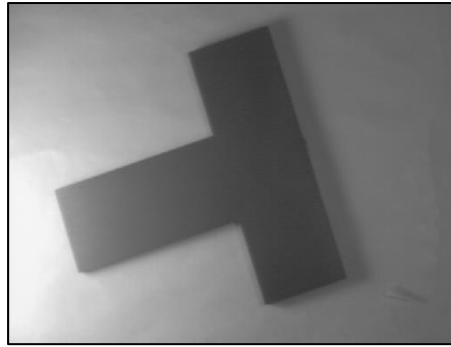
Multilevel thresholding

Thresholding

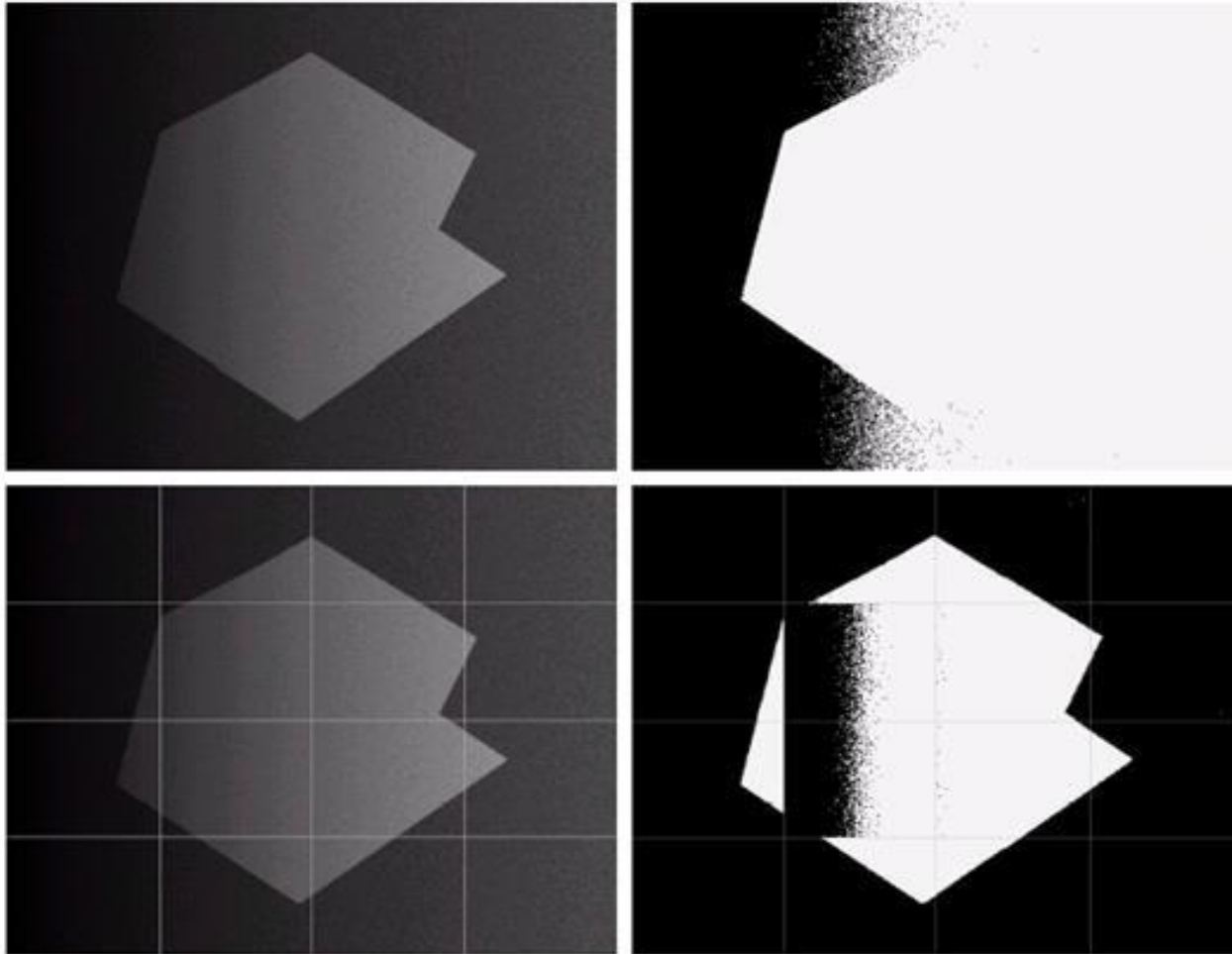
- Non-uniform illumination:



Global Thresholding



Adaptive Thresholding



Adaptive Thresholding

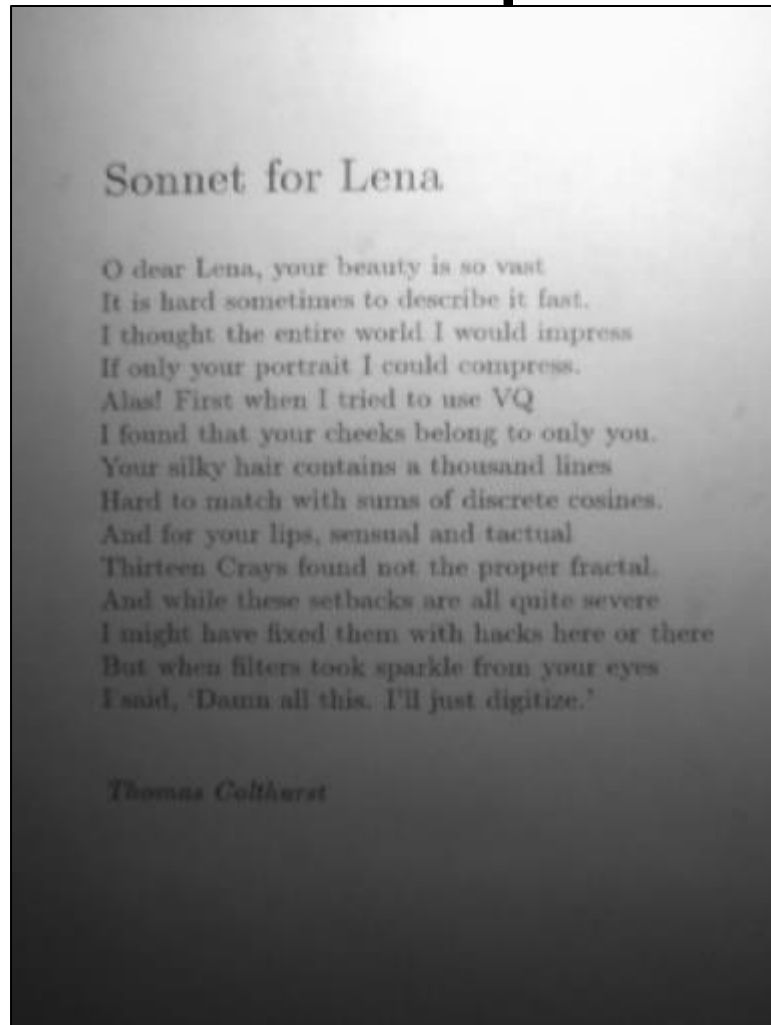
- Threshold: function of neighboring pixels

$$T = \textit{mean}$$

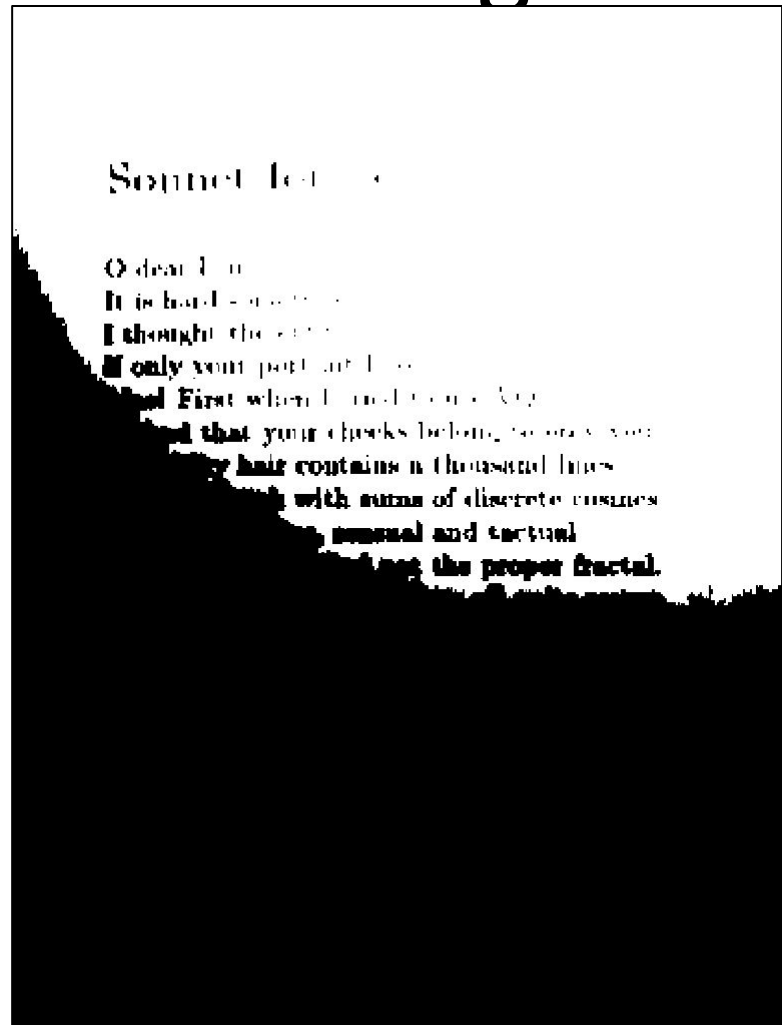
$$T = \textit{median}$$

$$T = \frac{\text{max} + \text{min}}{2}$$

Adaptive Thresholding

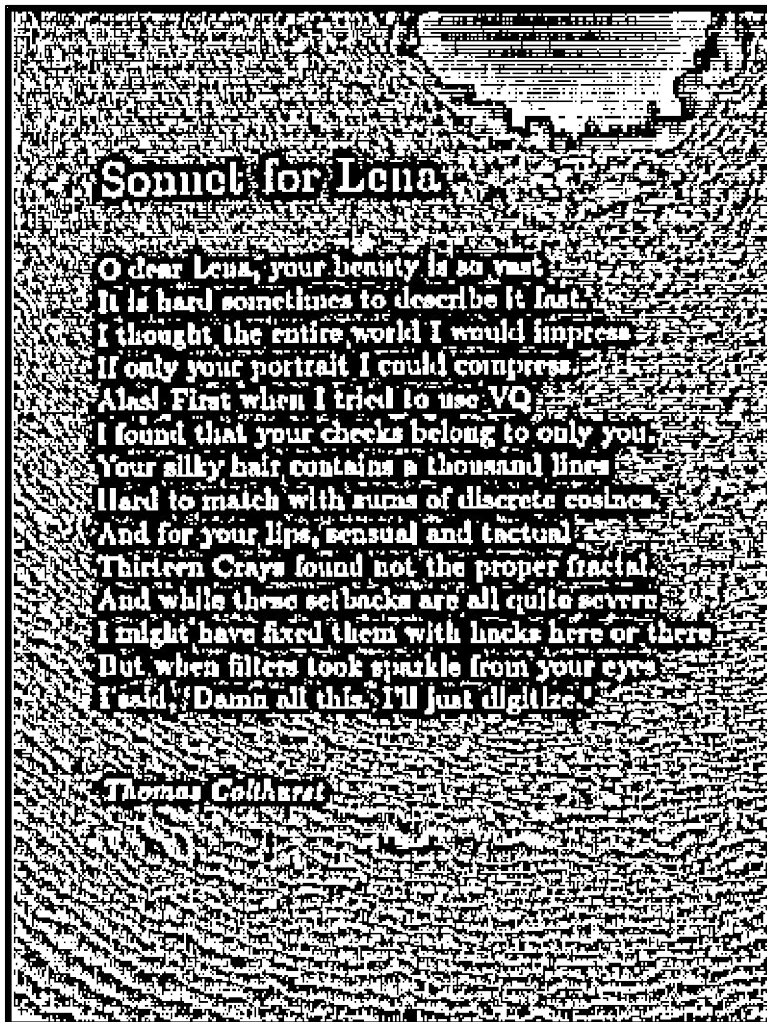


Original Image

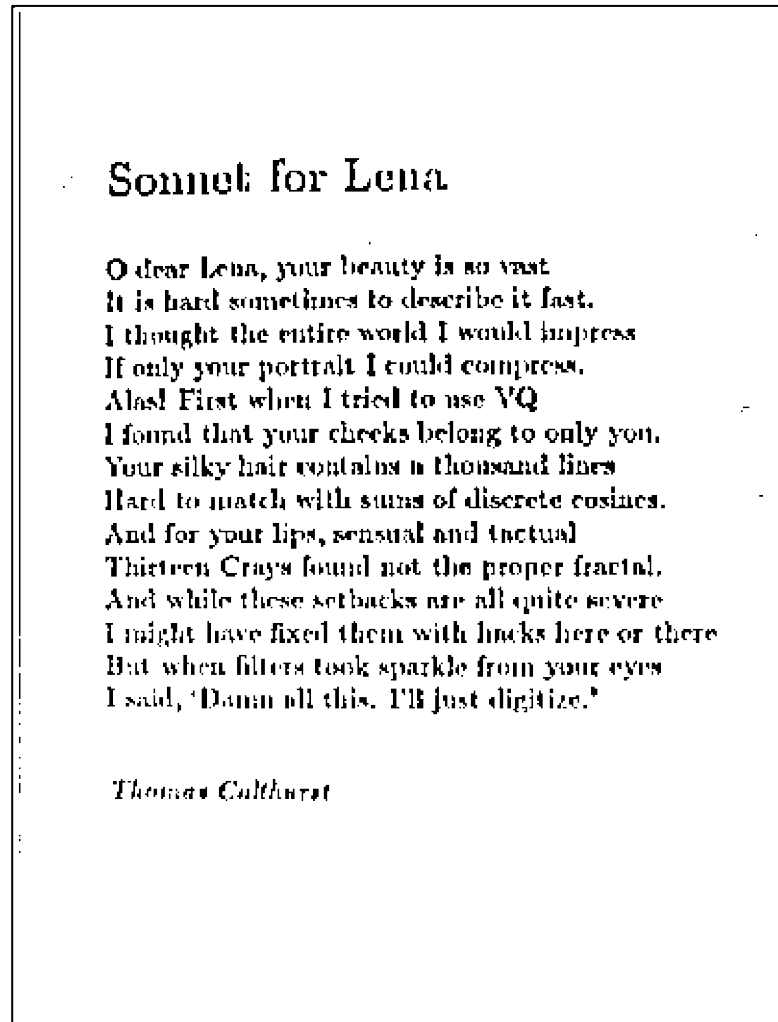


Global Thresholding

Adaptive Thresholding



T=mean, neighborhood=7x7



T=mean-Const., neighborhood=7x7

Adaptive Thresholding

- Niblack Algorithm

$$T = m + k \times$$

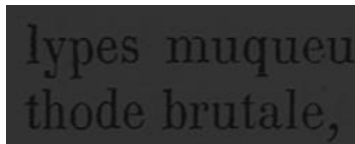
$$s \quad m = \text{mean}$$

$$s = \text{standard} \\ \text{deviations}$$

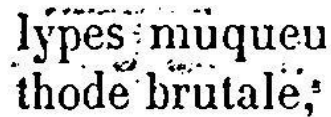
$$k = \quad \text{Niblack constant}$$

Document Binarization

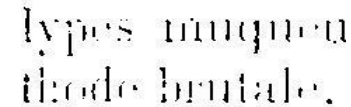
- Local Thresholding – Examples

The image shows the original document text, which is a scan of a document. The text is "lypes muqueu thode brutale," and it is presented in a dark, somewhat blurry font on a light background.

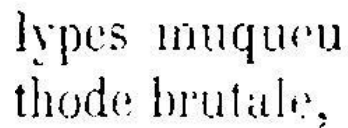
Original

The image shows the result of Niblack thresholding applied to the original text. The text is "lypes muqueu thode brutale," and it is presented in a high-contrast, black-and-white format where the background is white and the text is black.

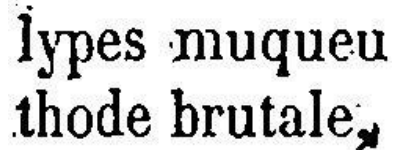
Niblack

The image shows the result of Sauvola thresholding applied to the original text. The text is "lypes muqueu thode brutale," and it is presented in a high-contrast, black-and-white format where the background is white and the text is black.

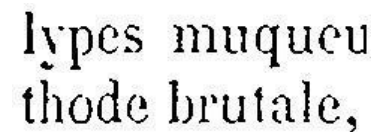
Sauvola

The image shows the result of Wolf thresholding applied to the original text. The text is "lypes muqueu thode brutale," and it is presented in a high-contrast, black-and-white format where the background is white and the text is black.

Wolf

The image shows the result of Feng thresholding applied to the original text. The text is "lypes muqueu thode brutale," and it is presented in a high-contrast, black-and-white format where the background is white and the text is black.

Feng

The image shows the result of NICK thresholding applied to the original text. The text is "lypes muqueu thode brutale," and it is presented in a high-contrast, black-and-white format where the background is white and the text is black.

NICK

Acknowledgements

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Computer Vision A modern Approach by Frosyth
- ◆ Digital image processing by Dr Usman Akram
- ◆ CSCI 1430: Introduction to Computer Vision by [James Tompkin](#)